Ming Gong, Ph.D.

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Summary

Hands-on **applied-AI engineering leader** who builds AI-powered automation systems and agents deployed to real operational workloads — for both internal-workforce and external customers. Currently Head of ML Engineering at Atlassian, the single point of accountability for an extended org of **130+** (60+ direct reports plus dotted-line partner teams) across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search Experiences. Previously Group Senior Principal Applied Sciences Manager at Microsoft, built and led a 40+ team that took Bing Question-Answering and Copilot Search from English-only to 100+ languages and 230+ markets, and shipped Copilot Search as a Bing AI vertical **one month ahead of Google's AI Mode**.

Operates across the full stack from frontier AI research through production serving: LLM post-training and SLM fine-tuning, agentic frameworks, knowledge graphs, retrieval, ranking, generative UI, and large-scale ML serving. Uniquely multi-dimensional — **applied sciences & full-stack engineering**, **AI products & research** (50+ publications, 11K+ citations, 20+ patents), **consumer & enterprise products**. 13+ years engineering, 7+ years engineering management, 6 years prior as an IC engineer / tech lead.

Selected Highlights — Aligned to Enterprise AI Transformation

- **Agentic AI & automation in production: Jira Agents in Rovo Chat** — launched within 3 months, a primary driver of Rovo adoption; code-driven agent architecture beat Manus and GenSpark on the **GAIA benchmark** in early 2026; **X-SuperAgent** (Atlassian's first OpenClaw-compliant agent framework) — **40%+** developer-productivity uplift. **Copilot Search** (Bing AI vertical) shipped April 2025, one month before Google's AI Mode.
- **0→1 builder in high-ambiguity spaces:** Jira Search MVP in **4 months**, full GA in 12 months → **3.4× Search MAU (2.5M → 8.5M)**; Enterprise Generative AI Search POC in **<1 month**, MVP GA in early 2026 — identified a cross-product platform gap no team was chartered to own; Bing QnA built from zero, placed the universal multilingual bet ∼1 year before it became official org strategy.
- **AI-native tooling & measurement at scale:** Established LLM post-training and SLM fine-tuning framework (incl. R1-distilled SLMs); **<5 s first-token latency** at production scale; **Glow** distilled SLM framework reduced first-token latency from 20–30 s to ∼**5 ms**; first BERT model deployed in Bing within 1 month of BERT open-source, zero additional GPU cost.
- **Knowledge systems & data infrastructure for enterprise:** Built Atlassian Knowledge Fundamentals — collaboration graph, entity linking, entity recommendation now adopted by multiple product teams; led entity-linking turnaround from **40% to 90%+ precision**. Knowledge Card on Bing — **150M entities, 60% query traffic, 65M DAU**; certified a Bing "WoW" feature (<10 out of 100+).
- **Hands-on leader at scale:** Single POC for an **extended org of 130+** at Atlassian (60+ direct + dotted-line partner teams); grew 30 → 60+ in one year; established greenfield Canada and Singapore sites (firsts for the company). Previously built a 40+ team at Microsoft from scratch across two sites.
- **Internal platforms & developer tools: Lumina** (Atlassian's generative UI framework — 20% engagement uplift, adopted as the org-wide GenAI standard); **Glow** (SLM serving framework); **Orca** (NLU platform powering 95%+ of Bing answers); **NeuronBlocks** (open-source NLP DNN framework, EMNLP/IJCNLP 2019); **Microsoft AI School APRD** (region-wide AI Talents Cultivation Platform).
- **Cross-functional influence in complex environments:** Bing global (100+ languages, 230+ markets); Atlassian cross-product enterprise; co-owned NRT inference pipeline with 8 platform teams across time zones; V-team of ∼30 → **115 contributors across 8 partner orgs** with a Coalition Council to drive cross-org alignment.
- **Research depth that ships:** Best Paper Award, IJCAI '24 AI4Research; **2× Microsoft MLADS Distinguished Contribution Awards** (first-ever back-to-back; CTO Office global awards); **50+ peer-reviewed papers, 11K+ citations, 20+ granted patents**.

Experience

Atlassian (US) — **Feb 2025 – Present**
Head of Machine Learning Engineering — Knowledge AI — *Bellevue, WA*

- **Agentic AI for the enterprise.** Launched **Jira Agents in Rovo Chat** within 3 months, becoming a primary driver of Rovo adoption. Code-driven agent architecture beat Manus and GenSpark on the **GAIA benchmark** in early 2026. Pioneered AI-native team operation by building **X-SuperAgent** (Atlassian's first OpenClaw-compliant agent framework) — boosting developer productivity by **40%+**.
- **Enterprise Generative AI Search, 0→GA.** Initiated and led from zero — identified a cross-product platform gap no team was chartered to own; POC in **<1 month**, MVP shipped in early 2026. The novel Generative UI layer (**Lumina**) — 20% engagement uplift, 30% SBS lift vs. competitor AI experiences — was adopted as Atlassian's standard generative presentation framework across all GenAI products and features. Previewed at **Team '25 EU AI Super Session**; live demo at **Team '26 Founder Keynote**.
- Generative Answers in search and AI Definitions drove sustained **AI MAU from 100K to 1.2M** via a novel generative UI framework.
- **Jira Search MVP in 4 months, full launch in 12.** Rebuilt ranking stack and search UX, driving **3.4× Search MAU (2.5M → 8.5M)** and helping AI Org exceed Search MAU L1 OKR; premiered at **Team '25**. Formed a V-team growing from ∼30 to **115 contributors across 8 partner orgs** with a Coalition Council for cross-org alignment.
- **Knowledge Fundamentals as a shared platform.** Built collaboration graph, entity linking, and entity recommendation — now adopted by multiple product teams for personalized features. Led entity-linking turnaround from **40% to 90%+ precision** — invested 2 months in hands-on diagnosis before acting, protected team stability through parallel senior hiring, then redirected technical approach.
- **Org leadership.** Single point of accountability for the Knowledge AI charter, leading an **extended org of 130+** — 60+ direct reports (70% scientists, 30% engineers; full-stack and globally distributed) plus dotted-line partner teams. Grew the team from 30 to 60+ in one year,

recruiting **2 Principal-level Managers and 3 Principal-level Tech Leads**; established new teams in Canada and Singapore (greenfield sites for the company).

Microsoft — Search Technology Center Asia (Bing, Edge, Copilot) **Aug 2013 – Dec 2024**
Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015) *Beijing / Suzhou*

- **Built and led an org of 40+ engineers and applied scientists from scratch** (70% scientists, 30% engineers; 5 managers — 3 science, 2 engineering) across Beijing and Suzhou, spanning Bing Search, Copilot, Cross-Lingual QA, Knowledge Graph, Recommendation, and Edge browser AI.
- **Copilot Search — Bing AI vertical (2024–2025).** Delivered a multi-turn AI search experience unifying GenQnA, GenSERP and Magazine; built scalable LLM generation pipelines and task-specific SLMs (incl. R1-distilled). Shipped April 2025 — **one month before Google's AI Mode**.
- **Generative SERP & the Glow SLM serving framework (2023–2024).** Reinvented Bing SERP as "AI Masterpieces" — proposed a tier-wise hallucination-control framework against split organizational consensus (conservative vs. velocity camps); held the position through 5–6 leadership reviews over two months. Framework adopted by Edge Copilot and Office Copilot. Built **Glow**, a distilled SLM framework replacing GPT-4-class models for production serving — first-token latency from 20–30 s down to $\sim$**5 ms**; V2 pipeline scaled to 5–10% query coverage.
- **Generative Answers on Bing SERP (2023–2024): 20% query coverage on Bing global markets, +2.0% WPSBS** via streaming generative answers and click-through to Bing Chat.
- **Bing Question-Answering: Zero-to-Global, Traditional ML $\rightarrow$ DL $\rightarrow$ Pre-trained Models $\rightarrow$ LLM (2016–2024).** Built Bing QnA from the ground up and led its evolution through every major AI paradigm shift. Placed the universal multilingual bet $\sim$1 year before globalization became an official org strategy. Scaled to **100+ languages and 230+ markets** via systematic cross-lingual methodology — **2$\times$ MLADS Distinguished Contribution Award** (first-ever back-to-back recipient; CTO Office global awards). Shipped 10+ DNN releases replacing the legacy GBDT ranker (400+ hand-crafted features); won WPSBS vs. Google. Co-owned NRT inference pipeline with 8 platform teams across time zones, treating partners as co-owners with shared credit — pipeline became Platform team's flagship demo and blueprint for subsequent products. Authored novel model compression powering the first BERT model deployed in Bing (within 1 month of BERT open-source, zero additional GPU cost) — **+3.0% coverage, +5.0% precision**. Universal model served 100+ languages on $\sim$**13%** of the GPU footprint a per-language approach would have required.
- **Knowledge Card for 150M Entities (2021–2023).** Multi-modal entity aggregation across web sources via template + MRC DL extraction and LLM generation. **60% Bing query traffic, 65M DAU, +3.0% WPSBS**; certified as a Bing "WoW" feature (<10 out of 100+ features).
- **Microsoft Copilot on Edge (2023–2024):** long-document summarization and chat over PDFs and long-form web pages — 1000+ page docs at **95%+ accuracy**; demoed at the Microsoft Copilot 2023 announcement event; shipped to Edge and Windows Copilot.
- **NLU Platform "Orca" (2020–2025):** unified query understanding pipeline (domain, intent, slots, topics) powering answer triggering. **95%+ of Bing answers** (coverage $\geq$ 0.05%) onboarded; 60% of onboarding answers closed triggering gap with Google.
- **Query-Dependent Passage Extraction (2019–2022):** replaced industry-standard sliding-window passage indexing with runtime query-dependent extraction via GNN document modeling + multi-teacher distillation to 40 ms latency. **Held #1 on Google Natural Questions benchmark for $\sim$1 year**; architecture adopted across Bing QnA, Knowledge, People Also Ask, and Azure Custom QnA.
- **Segment Entity Search & Recommendation (2021–2024):** shipped recipe, job, and movie answers — beat Google on relevance and WPSBS on recipe and job; closed WPSBS gap with Google on movie recommendation.
- **Underside on Edge — Right-Rail Insight Pane (2022–2024):** built the AI-powered right-rail panel for Edge featuring page summary, topics, QnA, related search, and related pages (feeds). **Edge engaged DAU +10%, Search DAU +1.2M.**

Platforms & Internal Tooling Built

- **Lumina** (Atlassian, 2025–2026) — generative UI framework powering Enterprise GenAI Search; **20% engagement uplift, 30% SBS lift** vs. competitor AI experiences; adopted as Atlassian's standard generative presentation framework across all GenAI products.
- **X-SuperAgent** (Atlassian, 2025–2026) — Atlassian's first OpenClaw-compliant agent framework for AI-native team operation; **40%+** developer-productivity uplift.
- **Glow** (Microsoft, 2023–2024) — distilled SLM serving framework replacing GPT-4-class models for production; first-token latency **20–30 s $\rightarrow$ $\sim$5 ms**; V2 scaled to 5–10% Bing query coverage.
- **Orca** (Microsoft, 2020–2025) — unified NLU platform (domain, intent, slots, topics) powering answer triggering for **95%+ of Bing answers**.
- **NeuronBlocks** (Microsoft, 2019, open source) — "Build NLP DNN models like playing Lego." Published at EMNLP/IJCNLP 2019.

People & Org Development

- **Founder & Head, Microsoft AI School — Asia-Pacific R&D Group** (2017–2024). Microsoft's region-wide AI Talents Cultivation Platform; led a **50+ operation v-team**; drove planning, curriculum, and execution of all AI learning events for the region. **10K+ participants, 2K+ honored graduates**, many of whom grew into AI talents and leaders across the Microsoft APRD org; record-breaking participation 8 years running — an early example of AI-augmented hybrid-workforce learning at scale.
- **Invited instructor, LinkedIn Global Academy for Chinese Programmers** (2022–2023): 4-class series on "Intelligent QA Systems for Search Engines" — **50M impressions, 98% satisfaction**.
- **Invited speaker, InfoQ AICon Global AI Conference** (2021): represented Microsoft China on globalization of Bing QA.

Education

Ph.D., Graphics and Visual Computing 2008 – 2013
Institute of Computing Technology, Chinese Academy of Sciences

Selected Awards

- **Best Paper Award**, IJCAI '24 AI4Research — *"Step-Back Profiling: Distilling User History for Personalized Scientific Writing."*
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+). Broke conference history to receive the award twice consecutively.
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+).
- 2023 Microsoft Global Hackathon — Edge Theatre, Hack for Web Experiences: **Best Overall Hack**; Greater China Region (Beijing): **Most Popular Hack**.
- 2023 Microsoft Global Hackathon — WebXT China, DeKG (decentralized knowledge ecosystem): **Best Innovativeness & Originality Hack**; Beijing Outstanding Project Award.
- Excellent Operation Award of AI School China — 2018, 2019, 2020, 2021, 2022, 2023.
- Microsoft AI Core Q3 FY18 **Greatness Award** (2018).

Selected Publications (50+ papers, 11,000+ citations on Google Scholar — full list on scholar profile)

▷ LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. "TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs." *Intelligent Computing* 2023.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. "Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers." *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. "Large Language Models are Diverse Role-Players for Summarization Evaluation." *NLPCC* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. "Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training." *AAAI* 2019.
- **Ming Gong**, Linjun Shou, Wutao Lin, et al. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
- Junwei Liao, Yu Shi, **Ming Gong**, et al. "Generating Human Readable Transcript for Automatic Speech Recognition with Pre-trained Language Model." *ICASSP* 2021.

▷ Code Intelligence (Developer Tools)

- Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, **Ming Gong**, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, Ming Zhou. "CodeBERT: A Pre-Trained Model for Programming and Natural Languages." *EMNLP (Findings)* 2020.
- Junjie Huang, Duyu Tang, Linjun Shou, **Ming Gong**, et al. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

▷ Model Compression, Knowledge Distillation & Efficient Serving

- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. "Reinforced Multi-Teacher Selection for Knowledge Distillation." *AAAI* 2021.

▷ Knowledge Graphs, Entity Understanding & Information Extraction

- Zilin Xiao, **Ming Gong**, Jie Wu, et al. "Instructed Language Models with Retrievers Are Powerful Entity Linkers." *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. "Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency." *Findings of EMNLP* 2023.
- Zimeng Li, Bo Shao, Linjun Shou, **Ming Gong**, et al. "WIERT: Web Information Extraction via Render Tree." *AAAI* 2023.
- Changzhi Sun, Yeyun Gong, Yuanbin Wu, **Ming Gong**, et al. "Joint Type Inference on Entities and Relations via Graph Convolutional Networks." *ACL* 2019.

▷ Search Relevance, Retrieval & Question Answering

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval." *ICLR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Multi-View Document Representation Learning for Open-Domain Dense Retrieval." *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.
- Xuguang Wang, Linjun Shou, **Ming Gong**, Nan Duan, Daxin Jiang. "No Answer is Better Than Wrong Answer: A Reflection Model for Document Level Machine Reading Comprehension." *EMNLP* 2020.

▷ Recommendation & Personalization

- Ning Wu, **Ming Gong**, Linjun Shou, Jian Pei, Daxin Jiang. "RUEL: Retrieval-Augmented User Representation with Edge Browser Logs for Sequential Recommendation." *CIKM* 2023.

- Hanshuang Tong, Jun Li, Ning Wu, **Ming Gong**, et al. "Ploutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model." *arXiv* 2024.

▷ **Cross-Lingual NLU & Multilingual Scaling**

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. "From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension." *AAAI* 2022.
- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.

Selected Patents (20+ granted; selection below)

- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022. (MS#412496-CN-NP)
- Sentence Representation Generation for Cross-lingual Retrieval. 2022.
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)